

Make your own app using Google App Engine

>>This workshop shows you how to build a registration mechanism that'll use a Google account to authenticate potential users> by Ashesh Lal

In this current era of modern web technologies and application development, Google has come out with yet another project – Google App Engine. The App Engine allows developers to make web applications that run on the appspot web server and uses Python, Java and Go as the supported programming languages while php is still in the experimental phase. The WSGI (Web Server Gateway Interface) is a standard used for web application development. Web application frameworks such as webapp2 handle these details for you, so you can focus your development efforts on your application's features. Google App Engine supports any framework written in pure Python that speaks WSGI, including Django (www.djangoproject.com), CherryPy (www.cherrypy.org), Pylons (www.pylonsproject.org), web.py (<http://webpy.org>), and web2py (www.web2py.com).

Getting Started

Before you begin with this workshop, make sure you have the following:

1. A Google account – gmail or appspot will do
2. A mobile number for administrator authentication
3. The App Engine SDK for python that can be found at <http://digit.in/dappengine> along with Python 2.7 SDK that can be found at <http://digit.in/pythondnld> or you can look for them in our Omega DVD.
4. A basic knowledge of Python and a Python editor. We used Sublime Text 2, which can also be found on the DVD.
5. A basic knowledge of HTML and character escaping.

Let's Begin!

1. Go to <https://appengine.google.com/>
2. Click on Create Application. Once you have found an available Application Identifier(your-app-id), fill in the Application title and leave the authentication at default.
3. Submit by clicking on Create Application.
4. You now have a custom domain “your-app-id”@appspot.com

Programming with App Engine

For a demo, go to <http://digit-app.appspot.com>

This workshop will give you a basic gist about how to go about programming with App Engine using Python on the Windows Platform. This will be done by creating an application that uses the users google account to authenticate and sends a mail to confirm the same.

1. Launch the Google App Engine
2. Go to Edit > Preferences

3. Browse for the path of Python, App Engine and your preferred text editor. Go to File > Create New Application
4. After you have filled in your application name and browsed for your target destination, submit by clicking on Create. In the App Engine launcher, click on the Edit option. This will open your text editor with a sample ‘Hello, World!’ program in main.py file or atleast the app.yaml file will open. You can thereafter add the target folder by going to File > Open Folder and clicking OK.
5. Replace the contents of main.py with the following code:

```
import webapp2
from google.appengine.api import users
from google.appengine.api import mail
class MainHandler(webapp2.RequestHandler):
    def get(self):
        self.response.write('''
            <form action="/signup" method=post>
            <input type="submit" value="Sign Up"/>
            </form>
        ''')
class SignUp(webapp2.RequestHandler):
    def post(self):
        self.response.write('<h2>Hey there!</h2> <br><br>')
        user = users.get_current_user()
        if user:
            greeting = (''Welcome, %s!
                <br>
                (<a href="%s">Sign
out</a>) <a href="%s"> Not you?</a>
                <br>
                <h3>Please check your
e-mail for further information</h3>''')
                % (user.nickname(),
users.create_logout_url('/'), users.create_logout_
url('/'))
            message = mail.EmailMessage()
            message.sender = "Ashesh Lal
<ashesh.lal@gmail.com>"
            message.subject = "Welcome to
digit-app, %s" %user.nickname()
            message.to = user.email()
            message.body = ""
            You have completed the
registration process.
```

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